

1. Manhattan Distance (L1 Distance)

Formula

The Manhattan distance (or *taxicab distance*) between two points $x = (x_1, x_2, \dots, x_n)$ and $y = (y_1, y_2, \dots, y_n)$ in n -dimensional space is the sum of absolute differences along each dimension:

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

This is a special case of the Minkowski distance when $p = 1$.

Numeric Example

Consider two 2D points:

Point A: (1,3)

Point B: (4,7)

Calculation:

$$d = |1 - 4| + |3 - 7| = |-3| + |-4| = 3 + 4 = 7$$

Interpretation: The distance is 7 units, mimicking a path along grid lines (e.g., moving 3 units horizontally and 4 units vertically).

2. Minkowski Distance

Formula

The Minkowski distance generalizes Euclidean ($p = 2$) and Manhattan ($p = 1$) distances. For points x and y , it is defined as:

$$d(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

Key cases:

$p = 1$: Manhattan distance.

$p = 2$: Euclidean distance.

$p \infty$: Chebyshev distance (maximum dimension difference).

Numeric Example

Using the same points $A(1,3)$ and $B(4,7)$, compute for $p = 3$:

1. Compute absolute differences raised to $p = 3$:

$$|1 - 4|^3 = 3^3 = 27, |3 - 7|^3 = 4^3 = 64$$

2. Sum and take the p -th root:

$$d = (27 + 64)^{\frac{1}{3}} = 91^{\frac{1}{3}} \approx 4.50$$

Comparison:

$p = 1$: Distance = 7 (Manhattan).

$p = 2$: Distance = $\sqrt{3^2 + 4^2} = 5$ (Euclidean).

$p = 3$: Distance ≈ 4.50 .

3. Application in KNN

Parameter tuning: In scikit-learn, use `metric="minkowski"` and set `p` to switch between distance metrics (e.g., `p=1` for Manhattan, `p=2` for Euclidean).

When to use which:

Manhattan ($p = 1$): Robust for high-dimensional data or outliers; scales better than Euclidean distance when features have varying units.

Minkowski ($p \geq 3$): Emphasizes larger dimension differences (e.g., $|x_i - y_i|^p$ amplifies disparities). Optimal p is data-dependent—use cross-validation to select.

Best practice: Always normalize features so no single dimension dominates distance calculations.

Summary: Manhattan distance sums absolute differences, while Minkowski generalizes it with a power parameter p . In KNN, choose p based on data structure and problem context.